

A Hand-Drawn Diagram Is a Local Minimum of Crossings and Overlap, and Nowhere Near One of Edge Length

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Abstract

Automatic layout tools place the boxes of a diagram by minimising a weighted sum of aesthetic criteria: few edge crossings, no overlapping boxes, edges of similar length. If people drew the same way, a diagram laid out by hand would be a local minimum of that sum and a tool started from it would find nothing to change. We test each criterion separately on 1,571 diagrams whose coordinates a person chose: 571 WikiPathways biological pathways, 701 Reactome SBGN maps and 299 BPMN process models, three communities with different authoring tools. The test is direct: move each box a small distance in each of sixteen directions and ask whether any move lowers the criterion. Under crossings and overlap together, every box in a hand layout is held in two corpora and 94% are in the third ^P: the hand layout is a local minimum. Under uniform edge length, the criterion that carries 94% of the energy as tools weight it, no box is held^P. A stress layout of the same graphs by **neato** has the opposite profile: held under edge length, free under crossings and overlap. Fitting the weights by inverse optimisation returns weight zero for edge length on every corpus, which this explains: any positive weight on it makes the hand layout non-stationary. Of three further criteria, alignment of boxes into shared rows and columns holds half the boxes in pathways and three quarters in process models^P and almost none in the tool's layouts; orthogonality of edges holds fewer; separation of boxes from edges holds none. The diagrams people draw are near-trees (median edges per box 1.06, 0.96 and 1.03, against 1.30 for the Rome benchmark), so crossings, the criterion the field's theory is built on, have almost nothing to remove. A tool builder has two changes to make: stop spending the energy on edge length, and add alignment. The measurement runs on a small C library for stochastic search whose distinctive feature is a matched-budget random control, the latest of a series of tools by the author that report when their own answer should not be trusted.

1 Introduction

A diagram is a graph with a drawing: boxes for the nodes, lines for the edges, and a position for every box. When a program chooses the positions it minimises an *energy*, a weighted sum of quantities the field has agreed are bad: edges that cross, boxes that overlap, edges of very unequal length, boxes that sit on edges they do not belong to [3, 2, 4, 5]. The weights are asserted by the tool's author. The premise is that a person drawing by hand is doing the same thing more slowly, so that the drawing they accept is one a tool would also accept.

The premise has a direct test, one criterion at a time. Take a diagram a person drew. Move one box a small distance in each of several directions and ask whether any of the moves lowers

*Independent researcher. Code, parsed corpora and scripts: <https://github.com/Anode1/cjitter>, directory `example/diagrams`.

the criterion. If none does, the box is *held*: the person has put it where the criterion wants it. Do this for every box and the fraction held says how much of the layout is a local minimum of that criterion. A criterion that holds every box is one the person satisfied; one that holds no box is one the person did not consider, or traded away.

We ran the test on 1,571 diagrams from three unrelated communities (Section 4) under a pre-registered analysis (`PREREGISTRATION-STATIONARITY.md` in the repository). The findings, with the exploratory numbers:

1. Under crossings and overlap together the hand layout is a local minimum: the fraction of boxes held is 1.00, 1.00 and 0.94^P on the three corpora. Under uniform edge length it is not: 0.00, 0.06 and 0.10^P. Under the three criteria summed with the usual equal weights the length term dominates and 0.03 to 0.10^P are held, which is what an earlier version of this study, using a capped descent rather than the directional test, reported as “hand layouts are not stationary”. They are not stationary in edge length, and in nothing else among the base criteria.
2. A stress layout of the same graphs with the same box sizes (`neato`) is the mirror image: held under edge length on 0.26 to 0.47^P of boxes, held under crossings and overlap on only 0.51 to 0.84^P.
3. Fitting the energy’s weights by inverse optimisation gives uniform edge length the weight 0.00 on every corpus. Finding 1 says why: any positive weight on a criterion that holds no box makes the sum hold no box.
4. Of three candidate criteria not in the base energy, alignment of boxes into shared rows and columns holds 0.53, 0.28 and 0.79^P of boxes in hand layouts and under 0.06 in `neato`’s; orthogonality of edges holds 0.19, 0.13 and 0.45^P; separation of boxes from edges they do not join holds none. In WikiPathways a third of boxes share an exact coordinate with another box while only 2% of coordinates are integers, so the alignment is the person’s act and not the editor’s grid; in BPMN it is partly the grid and mostly the lanes.
5. The diagrams people draw are near-trees: median edges per box 1.06, 0.96 and 1.03, against 1.30 in the Rome benchmark the field tests on [16]. Crossings are what the theory is about [1]; on the diagrams people actually draw there are almost none to remove, which is part of why finding 1 holds.

The rest of the paper is the foundation behind the test (Section 2), the instrument (Section 3), the corpora, the method, the measurements, the alternatives not yet excluded (Section 7), and what is already known (Section 8).

2 Foundations

The energy. A layout of a graph with N nodes is a point $x \in \mathbb{R}^{2N}$, two coordinates per node, the boxes’ sizes held fixed. An aesthetic energy is

$$E_w(x) = \sum_{k=1}^K w_k T_k(x), \quad w_k \geq 0, \quad (1)$$

where each term T_k is a quantity to be made small. The tool’s author asserts the w_k ; the field of multicriteria layout is the choice of the T_k and the w_k [2, 4].

Local minimum. A point x is a local minimum of E if there is a radius $r > 0$ with $E(y) \geq E(x)$ for every y within distance r of x . For a differentiable E this implies $\nabla E(x) = 0$. Two of our terms are not differentiable: crossings is a count, piecewise constant in x , and overlap has corners where boxes touch. So the test uses no gradient. Fix a radius d and a set V of unit directions in the plane (sixteen, equally spaced). For node i and direction v let $x + dv^{(i)}$ be the layout with node i moved by d along v and every other node fixed. Node i is *held* at radius d if

$$E(x + dv^{(i)}) \geq E(x) \quad \text{for every } v \in V, \quad (2)$$

and the estimand is

$$q_d(x) = \frac{1}{N} \#\{i : \text{node } i \text{ is held at radius } d\}. \quad (3)$$

$q_d = 1$ says the layout is a local minimum of E over single-node moves of length d ; for smooth E and small d it says $\nabla E(x) = 0$ to within the resolution of V . Because (2) is a comparison of two energies, it is defined for every term, and because it is evaluated per term it separates the criteria a person satisfied from those they did not. A descent-based measure (how far a capped descent moves the layout, the estimand of the first version of this study) answers the same question for the sum but cannot separate the terms; it is retained as a secondary.

The inverse problem. Given layouts $x^{(1)}, \dots, x^{(G)}$ that people accepted, inverse optimisation asks for the weights w under which they are nearest to stationary [18, 19]. With the directional differences $D_{iv}T_k = T_k(x + dv^{(i)}) - T_k(x)$ in hand, node i is held under w exactly when $\sum_k w_k D_{iv}T_k \geq 0$ for all v , a set of linear inequalities in w . The residual is the smallest total violation over w on the simplex, a linear program, and it is zero only if some weighting of the terms makes the layout a local minimum. A term with $D_{iv}T_k < 0$ for some v at almost every node (uniform edge length, at any layout whose edges are not all the same length) cannot take positive weight without violating the inequalities at those nodes, which is the zero weight the random-search fit found.

The null and the control. A test that can only succeed is not a test. The *stationarity reference*: q_d on a layout a descent has converged on must be near one, and it is (0.94 to 1.00^p on every corpus and energy). The *tool control*: q_d on a layout produced by a real tool from the same graph, which says what a layout that does minimise something scores on each term. The *matched-budget control*: uniform random placement at the same number of energy evaluations, which the hand layout must beat, or “held under crossings” would be a property of sparse graphs and not of the person.

3 The instrument

The descents, the control and the paired statistics come from `cjitter`, a C library of four stochastic searches over a box of real variables: hill climbing with restarts, simulated annealing, a genetic algorithm, and uniform sampling as the control. Every run is byte-reproducible from its seed on every platform, every method constant is a visible field, and `cjitter_compare` runs all four at the same budget on the same seeds and reports an exact paired sign test, Holm-corrected, against the control. “Better than uniform sampling at equal cost” is the only claim it lets a search make.

It is the latest of a series of small tools by the author, each of which reports the condition under which its own answer is wrong:

year	tool	what it reports against itself
1997	backpropagation thesis [32]	descent on an error energy, derived
2001	label placer	per-label random steps, kept if the summed overlap fell
2011	least squares, now <code>linearrr</code>	when a line is the wrong shape
2025	<code>bpnn</code>	when the network has memorised the rows
2026	<code>cjitter</code>	when a search does no better than sampling

The 1997 thesis derives backpropagation as gradient descent on an error surface; the 2001 placer, deployed in industry, is the same descent without a gradient, one label moved a random step and the step kept when the total cost fell. That operator, a proposal that moves one object and is judged on the summed energy, is the `block` field of the library, and the directional test of equation (2) is the same proposal made exhaustive: every direction tried, none accepted, the answer read off. The present paper turns the instrument around: instead of descending an energy to place objects, it starts from where a person placed them and reads which terms pull.

4 The corpora

Three collections of diagrams whose node coordinates a person chose, from three communities with different authoring tools, all redistributable.

corpus	licence	files	parsed	15–40 nodes	median m/n
WikiPathways GPML, <i>Homo sapiens</i>	CC0	1016	571	306	1.06
Reactome SBGN-ML	CC0	1382	701	180	0.96
BPMN Academic Initiative	academic	[PENDING: n]	299	161	1.03

The layouts are working documents, not aesthetic submissions: Reactome diagrams are laid out by curators [27], WikiPathways pathways are drawn in PathVisio [28], and the BPMN models are students’ work in Signavio [29, 30]. Nobody was asked to draw nicely. The three tools share no layout code. Two WikiPathways files (WP5037, WP3391) carry coordinates converted from Reactome [31] and are dropped.

A diagram enters as its largest connected component; container glyphs (SBGN compartments, BPMN pools and lanes) are not nodes; SBGN process glyphs are nodes, so the Reactome graphs are bipartite and their m/n is the process graph’s; each layout is rescaled so that its bounding box fits the unit square, box sizes scaled with it. The pre-registered run restricts to 15 to 40 nodes; the pilot used 10 to 400. Duplicate pathways (ortholog copies of one human pathway) are removed by model identity. **[PENDING: counts after deduplication; the 41 to 100 node sensitivity]**

How the coordinates were made matters for the alignment criterion. Of the raw coordinates, 2% are integers in WikiPathways, 69% in Reactome and 71% in BPMN (46% multiples of five); the fraction of boxes sharing an exact x with another box is 0.33, 0.06 and 0.47, an exact y 0.28, 0.12 and 0.73. WikiPathways authors align by hand on a continuous canvas; BPMN authors work on a grid inside lanes; Reactome’s curation tool rounds but its authors align little.

5 The method

Terms. With n nodes, m edges, box i of width w_i and height h_i , L the median edge length of the layout under test:

term	definition	zero when
T_C crossings	pairs of edges with no shared endpoint that cross, $/m$	planar drawing
T_O overlap	total overlap area of boxes / total box area	no two boxes overlap
T_L length uniformity	mean over edges of $((\ell - L)/L)^2$	all edges of length L
T_R orthogonality	mean over edges of $\min(\Delta x , \Delta y)/(\Delta x + \Delta y)$	every edge axis-aligned
T_A alignment	mean over nodes of $1/(1 + \sum_{j \neq i} e^{-(\Delta x/s)^2} + e^{-(\Delta y/s)^2})$, $s = 0.02$	every node shares a row and column
T_N node-edge	penalty for a node within $(w + h)/4$ of an edge it does not join	no node sits on a foreign edge

The base energy is $T_C + T_O + T_L$ with weights $(1, 1, 1)$, the form of the simulated-annealing layout of Davidson and Harel [2] and of the multicriteria tradition since [24]. The field’s other default is stress [25, 26], $\sum_{i < j} (\|x_i - x_j\| - L \delta_{ij})^2 / \delta_{ij}^2$ with δ_{ij} the graph distance, which force-directed tools and **neato** minimise and which Dwyer et al. [12] found human choices track; it is the second base energy of the pre-registered run. **[PENDING: stress row in every table]** T_R , T_A and T_N are the candidates for what the base energy misses. The alignment definition above is the kernel form A3; two simpler forms, A1 (distance to the nearest node sharing a row or column) and A2 (rows and columns priced separately), are reported as sensitivities, together with HOLA’s gridiness [8], the fraction of nodes in some alignment of three or more, which is the published form. A1 has a corner at exact alignment and A3 does not, and the corner is what holds a box: A1 holds 0.53 of WikiPathways boxes and A3 holds 0.25%. Which form is primary is declared before the run.

All terms are evaluated here on straight centre-to-centre segments. GPML and SBGN store waypoints and BPMN routes edges orthogonally, so crossings, orthogonality and node-edge separation are measured on a drawing the person did not see. **[PENDING: the same terms on the stored polylines, with the fraction of edges that have waypoints per corpus]**

Directional test. Radius $d \in \{0.005, 0.01, 0.02, 0.05\}$ of the drawing width, sixteen directions, every node, each term alone and the declared sums. Primary cell $d = 0.02$. Deterministic: no seed, no budget.

Descent, secondary. Hill climbing, one node per proposal, initial step set from the cap, each node clipped to a disc of radius d around its start, 4000 energy evaluations, 15 seeds per instance; reported as the fraction of the cap used and the fraction of energy removed. The exploratory runs used a step of 0.1, five times the cap, so that nearly every proposal landed on the cap circle; the base-energy result did not depend on it (98.5 to 95.9% of the cap used as the step fell from 0.1 to 0.005) and the alignment result did (47 to 18%), which is why the directional test is primary.

Fitting. Weights on the simplex, the linear program of Section 2 on a training half of each corpus, residual reported on the held-out half; the random search of the pilot (200 log-uniform draws [6]) beside it as a check. One candidate term added to the base at a time, then all six.

Statistics. The unit of inference is the diagram. Per corpus, the paired comparison of q_d between two energies, or between the hand layout and the tool control, is the exact Wilcoxon signed-rank test with the exact sign test beside it, Hodges–Lehmann shift and distribution-free interval; Holm correction over the declared family within each corpus. Equivalence claims (“no reweighting repairs it”) are two one-sided tests against a declared margin, never a failure to reject. **[PENDING: the families, as declared in the signed pre-registration]**

6 Measurements

All numbers here are exploratory (one seed, 60 diagrams of 15 to 40 nodes per corpus for the directional test, 40 per split for the fits) and are replaced cell by cell by the pre-registered run. **[PENDING: confirmatory tables]**

Which criteria hold the hand layout. $q_{0.02}$, the fraction of boxes with no improving direction among sixteen, on the hand layout, on the same layout after an uncapped descent under the same energy (the stationarity reference), and on **neato**’s layout of the same graph with the same box sizes (the tool control).

corpus	energy	hand ^P	converged ^P	neato ^P
WikiPathways	crossings + overlap	1.00	1.00	0.55
	length only	0.00		0.26
	(1, 1, 1)	0.03	0.96	0.19
	crossings + overlap + alignment A1	0.53	1.00	0.06
	crossings + overlap + orthogonality	0.19		0.00
Reactome	crossings + overlap	1.00	1.00	0.84
	length only	0.06		0.47
	(1, 1, 1)	0.06	0.97	0.43
	crossings + overlap + alignment A1	0.28	1.00	0.05
	crossings + overlap + orthogonality	0.13		0.01
BPMN	crossings + overlap	0.94	1.00	0.51
	length only	0.10		0.46
	(1, 1, 1)	0.10	1.00	0.31
	crossings + overlap + alignment A1	0.79	1.00	0.05
	crossings + overlap + orthogonality	0.45		0.00

At the hand layout the three base terms share the energy as length 0.94, crossings 0.06, overlap 0.00. The converged column is near one everywhere the descent converged within its budget, so the test has power; the **neato** column is a layout that is a minimum of something else, and it reads as one.

Chance. The hand layout beats uniform random placement at the same evaluation budget on 350 of 373^P pathways, and descent from the hand layout ends lower than descent from a random start on 313 of 373^P. Being held under crossings is not a property of sparse graphs alone.

Reweightings. Held-out fraction of the cap a capped descent still uses (the secondary estimand; 0 would be stationary), from the pilot’s random-search fit:

energy	WikiPathways	Reactome	BPMN
asserted (1, 1, 1) ^P	98.7	91.7	92.8
base three, fitted ^P	99.0	91.7	90.4
+ node-edge separation ^P	98.0	91.3	93.0
+ orthogonality ^P	88.5	85.8	65.1
+ alignment (A1) ^P	61.0	79.9	46.9
all six ^P	57.2	79.7	58.4

Uniform edge length took weight 0.00 in every fit. The first table says why, and it also says the residual in this table is the length term’s: remove it and the base energy holds the layout outright. **[PENDING: the linear-program residual, per corpus, with and without T_L]**

7 What else could produce this

1. **The editor snapped to a grid**, so alignment is the tool’s. WikiPathways has 2% integer coordinates and a third of boxes exactly aligned, so there it is the person’s. BPMN is snapped. Test, declared: jitter every box by half a grid cell and repeat; the alignment that survives is the person’s.
2. **Straight segments where the person drew polylines**. GPML and SBGN store waypoints and BPMN routes edges orthogonally; crossings, orthogonality and node–edge separation are here evaluated on centre-to-centre segments the person never saw. Test: the same table on the drawn polylines. **[PENDING:]**
3. **The person minimised a constrained energy whose constraints are absent**: reading direction, layers, compartments, lanes. The BPMN y -alignment of 0.73 is the lanes. Test: a flow term (penalty on edges running against the reading direction) and containment as a hard constraint, added as candidates. **[PENDING:]**
4. **The radius**. At $d = 0.02$ on a thousand-pixel canvas the move is twenty pixels, about one grid step; a person cannot be stationary below the resolution they control. Test: the table at every cap, and at d in units of box width and of median edge length.
5. **Selection**. Near-trees make crossings easy. Test: q_d against m/n within each corpus.

8 Related work

Aesthetic criteria and their weighting go back to Tamassia, Di Battista and Batini [3] and were made an explicit energy by Davidson and Harel [2]; Purchase [4] gave the criteria their metrics, and that list has no alignment metric. Whether people agree with the criteria has been asked three ways. By having people draw: van Ham and Rogowitz [11] found hand layouts with fewer crossings than the force-directed algorithm produced; Dwyer et al. [12] had 32 participants lay out one graph and 194 judges rank the results against three automatic layouts, and found the judges’ choices track stress; Purchase, Pilcher and Plimmer [20] had participants draw from adjacency lists and found crossing removal the strongest aesthetic and alignment to a grid important; Kieffer et al. [8] collected 17 designers’ layouts of 8 small graphs, ranked by 66 judges, and found their gridiness, the fraction of nodes in some alignment of three or more, correlated with rank on all 8 graphs, which is where HOLA’s alignment step comes from. Alignment as a human criterion is therefore known since 2012; what is new here is the count of how many hand-placed boxes it holds, on diagrams of 15 to 40 boxes, beside the counts for the criteria tools weight. By showing people drawings: Purchase et al. [13] ran a Turing test in which hand drawings were distinguishable from algorithmic ones and judged better; Klammler, Mchedlidze and Pak [9] fitted the weights of a linear aesthetic combination by Nelder–Mead to discriminate good from worsened layouts, without human coordinates, and wrote that human-labelled data “may become a critically important accomplishment in the graph drawing field”. By learning from examples: Masui [10] learned a weighted constraint combination from user-supplied good and bad layouts by genetic programming, and Spönemann, Duderstadt and von Hanxleden [21] adjusted weights from user selections among candidates; neither tested whether the examples were minima of the learned energy. The closest prior descent from a hand layout is Flud [14]: simulated annealing continued from crowd layouts of pathway diagrams raised an asserted five-term score, on three networks, read as a success of the hybrid; the cap, the per-term test, the reference and the controls here are what turn that observation into a measurement of the score. Van Wageningen, Mchedlidze and Telea [15] show drawings with identical metric values that differ in quality, which is why a fitted energy with no residual would still not be the energy people use, and why the claims here stay on the negative side. Kobourov, Pupyrev and Saket [22] found crossings matter

little on sparse large graphs, consistent with finding 5. Inverse optimisation, the recovery of an objective from solutions taken as optimal, is due to Ahuja and Orlin [17] for linear programs and to Keshavarz, Wang and Boyd [18] for the convex case; Chan, Lee and Terekhov [19] define the goodness-of-fit residual reported here. Matched-budget random controls are standard in hyperparameter search [6] and were prescribed as the baseline for genetic algorithms in 1995 [7]; we have not found them in layout. No paper fitting energy weights to human-placed coordinates was found in four searches of 2024 to 2026 work; the nearest, Zhang et al. [23], learns from 64,222 pairwise preference labels, not coordinates.

9 Reproducibility

Parsers, the energy (one function per term), the directional test, the descent, the fitting scripts, the parsed corpora and every table’s script are in the repository under `example/diagrams`. The directional test is deterministic; the descents are deterministic from their seed on every platform. Raw corpus archives are fetched by a script and checked by hash. **[PENDING: commit hash frozen for the run]**

10 Conclusion

People draw at a local minimum of the criteria that are either satisfied or not: no box on another, no crossing a small move removes. They draw nowhere near a minimum of uniform edge length, the smooth criterion that carries most of the energy in the tools, and a tool’s own layouts show the opposite profile. What people do that the base energy does not price is align boxes into rows and columns, by hand where the canvas is continuous. A tool builder reading this has two changes to make and one number to report: take the weight off edge length, add alignment, and print the fraction of boxes the energy holds when started from a layout a person accepted.

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